

ALPER ALIMOĞLU, PH.D.

Python Backend & Distributed Systems | Blockchain & Smart Contracts

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SUMMARY

Computer Engineering Ph.D. with 12+ years in software development, blockchain systems, and distributed computing. Combines academic research with industry experience; strong track record in Python backend development and delivery of production systems.

SKILLS

Python Flask REST APIs Solidity Bash Ethereum Smart Contracts GNU/Linux
Cloud Computing Blockchain Docker Distributed Computing

EXPERIENCE

Data Scientist @Odeabank Nov 2025-Present Istanbul, Turkey

Refactoring a Python backend project (~10k lines) that recommends optimal fund offers to customers; significantly improved run-time performance.

Python Developer @Huawei Mar 2025-Aug 2025 Istanbul, Turkey

Fixed bugs in database Python projects. Co-authored an academic paper (main author).

Part-time Lecturer @Boğaziçi University Feb 2025-Jun 2025 Istanbul, Turkey

Taught *Functional Programming and Verification* to senior undergraduate students.

Middleware Administrator @Anadolu Hayat Emeklilik Aug 2024-Nov 2024 Istanbul, Turkey

Developed a chatbot integrated into Microsoft Teams; backend development in pure Python Flask, using Microsoft's LLM; in use by all company members.

Cloud Engineer @ICRON A.Ş. Mar 2024-Apr 2024 Istanbul, Turkey

Responsible for the company's cloud on Azure; carried out database migration between regions.

R&D Engineer @Boğaziçi University Feb 2017-Oct 2021 Istanbul, Turkey

Python backend development for own Ph.D. research; core contributor to eBlocBroker (see Open-Source Projects).

Research Project Assistant @Research Foundation for SUNY Aug 2012-Jul 2015 NY, USA

Smart Load Balancer: Applied Echo State Network ML models for workload prediction in data centers.

b-ext4 File System: Hybrid SSD/HDD design to optimize performance and reduce energy consumption.

Internships @Cisco, Ericsson, ITU Robotics Lab 2009-2010 Istanbul, Turkey

Cisco University Prize-winning research intern. RoboCup 3D Simulation Team (ITU AI & Robotics Lab).

EDUCATION

Ph.D. Computer Engineering

Boğaziçi University

📅 2016-2024

📊 GPA 3.64/4.0

Thesis: **EBlocBroker: Blockchain-based Computational Broker for e-Science** Advisor: Prof. Can Ozturan

M.Sc. Computer Science

Binghamton University - SUNY

📅 2012-2015

📍 Valedictorian, GPA 3.77/4.0

Thesis: **Short-Term Workload Prediction with Reservoir Computing** Advisor: Prof. Kanad Ghose

B.Sc. Information Systems Engineering (Dual Diploma Program)

Binghamton University - SUNY & Istanbul Technical University

📅 2008-2012

📍 Summa Cum Laude, GPA 3.89/4.0 (SUNY), Overall 3.62/4.0

PUBLICATIONS

Journal Articles

- Alimoglu, A., Özturan, C. "**Autonomous Blockchain Organization for Sustainable Software and Execution History**", *Blockchain: Research and Applications* (ESCI Q1), Dec 2025 (accepted - in press).
- Alimoglu, A., Özturan, C. "**An Autonomous Blockchain-Based Workflow Execution Broker for e-Science**", *Cluster Computing*, Springer (SCI-E Q1), May 2024. DOI:10.1007/s10586-024-04534-z
- Alimoglu, A., Özturan, C. "**An Autonomous Blockchain-Based Computational Broker for e-Science**", *Concurrency and Computation: Practice and Experience*, Wiley (SCI-E Q1), Mar 2024. DOI:10.1002/cpe.8087

Conference Papers

- Alimoglu, A., Erdayandi, K., Mustafa, M. A., Cali, Ü. "**EDGChain-E: A Decentralized Git-Based Framework for Versioning Encrypted Energy Data**", *IEEE International Conference on Distributed Ledger Technologies (ICDLT)*, Dec 2025 (accepted - in press). arXiv:2507.01615
- Alimoglu, A., Özturan, C. "**Design of a Smart Contract Based Autonomous Organization for Sustainable Software**", *IEEE 13th International Conference on e-Science*, Oct 2017. IEEE Xplore Link
- Shahriar Asta, Tuna Sonmez, Onuralp Ulusoy, Alper Alimoglu, Mustafa Ersen, Ozdemir Sozmen, Sanem Sariel-Talay. "**BeeStanbul Team Description Paper**", *RoboCup Symposium Workshop*, Istanbul, Jul 2011. PDF Link

OPEN-SOURCE PROJECTS

eBlocBroker

github.com/ebloc/ebloc-broker

Developed from scratch in pure Python for Ph.D. thesis. Blockchain-based autonomous computational resource broker for e-Science: Solidity smart contracts, Python backend, IPFS integration. Job scheduling, data provenance, reproducibility, execution history.

Trade Bot

github.com/avatar-lavventura/trade_bot

Fun side project: Binance Spot trading bot in Flask. REST/WebSocket integration with exchange APIs, real-time market data, algorithmic strategies, autonomous execution. Fault-tolerant long-running Python backend.

HONORS & AWARDS



Valedictorian

Graduate commencement speaker, Binghamton University, 2014 , Speech: "Protect Our Dreams"



Summa Cum Laude

Top of Class, B.Sc. in Information Systems Engineering, 2012



Cisco University Prize

Winner, 2010



Dean's List

Multiple honors at Binghamton University - SUNY & ITU



Academic Achievement Award

Binghamton University - SUNY, GPA 3.98/4.0, 2012